

CIE-2 Syllabus

- **ADA:**

- Brute force-String matching, Exhaustive search-Knapsack problem, TSP, Job assignment problem
- Dynamic Programming-Coin problem-Coin row problem, change making problem, coin collection problem, Warshall's algorithm, Floyd's algorithm, Knapsack problem- Bottom up approach and top down approach using memory functions
- Decrease and Conquer- Three variations, Topological sorting using source removal method, Generating permutations-Johnson Trotter algorithm, Generating permutation using bottom up minimal change algorithm and in lexicographic order, Russian Peasant multiplication, Median selection problem using variable size decrease
- Transform and Conquer-Presorting-Binary search, Uniqueness of an element and mode value, Heap sort, Horner's rule
- Heap sort - Bottom up approach and Top down approach.

- **OS:**

- Unit 2: Multithreaded Programming: Overview, Multi-core Programming, Multithreading Models, Implicit Threading, Threading Issues. Thread Scheduling, Multiple-Processor Scheduling, Real-Time CPU Scheduling.
- Unit 3: Synchronization: Background, Critical Section Problem, Mutex locks, Semaphores, Classic Problems of Synchronization. Deadlocks: System Model, Deadlock characterization, Methods for handling deadlocks, Deadlock prevention, Deadlock avoidance, Deadlock Detection and Recovery from deadlock.

- **LAO:**

- Unit 4: eigen spaces of a linear transformation, Characteristic and Minimal Polynomials of Block Matrices, Jordan Canonical form.
- Unit 5: Diagonalization, Orthogonal diagonalization of real symmetric matrices, quadratic forms and its classifications, signature, index.

- **SE:**

- System models: Context models, Behavioural models, Data models, Object models, Structured methods.
- Architectural Design: Architectural Design Decisions, System organization Modular Decomposition styles, Control styles
- Object oriented design: Objects and Object Classes, An object-oriented design process Design evolution.
- Software Project Management: The Management Spectrum, Product, process and project, The W5HH principle, Critical practices, Estimation for Software Project: Software Project estimation, Decomposition Techniques, Empirical Estimation models

- **CRP:** unit 2 and 3 full

- **TFC** (as of 14-05-2026):

- Unit 2: Proving Languages Not to Be Regular, Closure Properties of Regular Languages, Equivalence and Minimization of Automata
- Unit 3: Context Free Grammars, Parse trees, Applications of Context Free Grammars, Ambiguity in Grammars and Languages, Eliminating Useless Symbols, Computing the Generating and Reachable Symbols, Eliminating Epsilon Productions, Eliminating Unit Productions, Chomsky Normal Form, Greibach Normal form